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Fluid Management for Dehydration

Oral Rehydration

- The American College of Physicians recommends reduced osmolarity oral rehydration solution (ORS) as first-line therapy for mild to moderate dehydration in adults, with a well-balanced ORS containing 65–70 mEq/L sodium and 75–90 mmol/L glucose [1,2]
- For mild diarrheal illness, diluted fruit juices, flavored soft drinks, saltine crackers, and broths may suffice, and ORS can be combined with other fluid types or supplemented with IV fluids when oral compliance is suboptimal [2]
- Nasogastric administration of ORS may be considered in patients with moderate dehydration who cannot tolerate oral intake, with a total fluid prescription typically ranging from 2200 to 4000 mL/day depending on ongoing losses [1,2]

Intravenous Rehydration

- The American Heart Association recommends isotonic crystalloids (0.9% normal saline or balanced salt solutions like lactated Ringer's) as the fluids of choice for severe dehydration, with an initial bolus of 15–20 mL/kg/hour (1–1.5 L in average adults) during the first hour in the absence of cardiac compromise [1,3]
- For patients with sepsis and hemodynamic instability, a 20 mL/kg bolus should be given initially, and rapid fluid administration should continue until clinical signs of hypovolemia improve (hypotension, low urine output, impaired mental status) [2,3]
- After initial resuscitation, fluid choice depends on corrected serum sodium, with normal or elevated corrected sodium requiring 0.45% NaCl at 4–14 mL/kg/hour, and low corrected sodium requiring 0.9% NaCl at similar rates [3]

Monitoring and Special Populations

- The National Kidney Foundation recommends urine output should exceed 0.5 mL/kg/hour, and fluid administration rate must exceed ongoing losses (urine output + 30–50 mL/hour insensible losses + gastrointestinal losses) [2,3]

- In elderly patients, monitor closely for fluid overload, particularly with cardiac or renal disease, and isotonic fluids should be administered with heightened caution for fluid overload [2,4]
- For diabetic ketoacidosis/hyperglycemic crisis, isotonic saline (0.9% NaCl) is the initial fluid of choice, infusing at 15–20 mL/kg/hour during the first hour, and subsequent fluid selection depends on corrected serum sodium and electrolyte levels [3]

Critical Pitfalls to Avoid

- Avoid rapid fluid resuscitation in mild to moderate hypovolemia, as it is unnecessary and potentially harmful [2]
- Do not administer aggressive fluid boluses in elderly patients with renal impairment to prevent acute pulmonary edema [4]
- Monitor for fluid overload in elderly patients, especially those with chronic heart or kidney failure, and ensure potassium replacement once renal function is confirmed (urine output ≥0.5 mL/kg/hour) [2,3,4]

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